

CS Arrays - Quick Notes for Placement

1. What is an Array?

An array is a data structure used to store multiple elements of the same data type in contiguous memory locations.

2. Why Arrays are Used?

Arrays help store and access multiple values using indexes efficiently.

3. Types of Arrays

1D Array, 2D Array (matrix), Multi-dimensional Array.

4. Advantages

Fast access using index, easy traversal, memory efficient for fixed-size data.

5. Disadvantages

Fixed size, insertion/deletion can be costly.

6. Time Complexity

Access: $O(1)$, Search: $O(n)$, Insertion: $O(n)$, Deletion: $O(n)$.

7. Important Array Interview Questions

Find maximum/minimum, reverse array, rotate array, find duplicates, prefix sum, two sum, kadane's algorithm, binary search.

8. Java Array Example

```
int[] arr = {10, 20, 30, 40};
```

9. Common Placement Topics

Sorting, searching, sliding window, two pointers, prefix sum, binary search on arrays.